

1. Smart Voice-Controlled Music & Dance Assistant (NLP-Based)

WHAT IT IS:

A voice-activated system that uses **Natural Language Processing (NLP)** to control music, lighting, and routines for dancing.

WHY I WANT IT:

- Makes dance practice easier and more fun
- No need to manually change songs or settings
- Helps me stay focused during practice
- Can act like a “virtual dance coach” by responding to commands

WHERE I WILL PLACE IT:

- Placed mainly in the **living room and kitchen** (high-traffic and mess-prone areas)
- Docking/charging station near the **kitchen wall**

HOW I WILL USE IT:

- I can say commands like:
 - “Play hip-hop dance playlist”
 - “Slow down the music”
 - “Turn on dance lights”
- The system processes my voice using NLP and performs the action
- It can also suggest routines or timers for practice sessions

2. Smart Pickleball Training Wall with ESP32 Sensors

WHAT IT IS:

A training wall using **ESP32** with motion and impact sensors that tracks ball hits and performance.

WHY I WANT IT:

- Helps improve my pickleball skills at home
- Tracks accuracy, speed, and number of hits
- Makes solo practice more interactive and competitive

WHERE I WILL PLACE IT:

- Installed at the **back deck** (ideal open space for practice)

HOW I WILL USE IT:

- Hit the ball against the wall
- ESP32 sensors detect impact and movement
- Data is sent to my phone (via WiFi/Bluetooth)
- I can see stats like:
 - Number of successful hits
 - Reaction time
 - Practice duration
- It can even gamify practice (score system or challenges)